

DDR基础课程

第五节 - 初步了解DRAM Jedec Spec 3

芯片马版权所有

目录

关注的问题

- 命令与波形
- 从用户的角度看如何读取一个数据回来
- 从用户的角度看如何写一个数据进去

命令

- 真值表

Table 198 — Allowable CAS command operand(s) combination

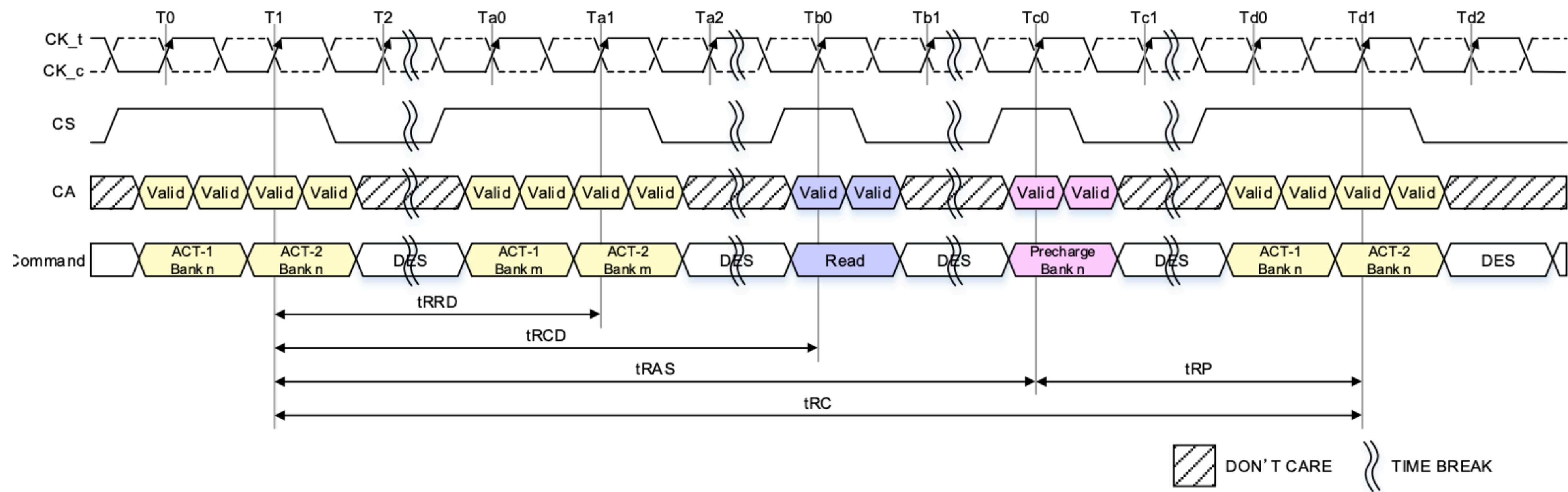
Operation Mode	CAS Command Operand(s)						
	WS_WR	WS_RD	WS_FS	DC0-3	WRX	WXSA	WXSBB3
WCK2CK SYNC (WR)	H	L	L	LLLL	L	L	L
WCK2CK SYNC (WR) + Write Data Copy	H	L	L	VVVV	L	L	L
WCK2CK SYNC (WR) + Write X	H	L	L	LLLL	H	V	V
WCK2CK SYNC (RD)	L	H	L	LLLL	L	L	V
WCK2CK FAST SYNC	L	L	H	LLLL	L	L	L
WCK2CK SYNC OFF	H	H	H	LLLL	L	L	L
Write Data Copy	L	L	L	VVVV	L	L	L
Write X	L	L	L	LLLL	H	V	V
Non-Zero Burst Start Address for RD	L	L	L	LLLL	L	L	H
WCK SUSPEND	H	L	H	LLLL	L	L	L
Illegal		Other Operands Combination					
NOTE 1 "V" means "H" or "L" (a defined logic level). NOTE 2 All CAS command operands are non-sticky. NOTE 3 Write Data Copy function requires DC0-3 operands not to be all-zero. NOTE 4 CAS-WCK_SUSPEND command is valid only when MR0 OP[2]=1B (Enhanced WCK Always On mode supported) and MR18 OP[4]= 1B (WCK Always On mode enabled)							

Table 197 — Command Truth Table (cont'd)

SDRAM COMMAND	BK ORG (BG, 16B, 8B)	SDR CMD PIN	DDR COMMAND PINS							CK_t edge	Notes
		CS	CA0	CA1	CA2	CA3	CA4	CA5	CA6		
CAS	Any	H	L	L	H	H	WS_WR	WS_RD	WS_FS	R1	1, 2, 7, 13
		X	DC0	DC1	DC2	DC3	WRX	WXSA	WXSB/B3	F1	
MULTI PURPOSE COMMAND (MPC)	Any	H	L	L	L	L	H	H	OP7	R1	1, 2
		X	OP0	OP1	OP2	OP3	OP4	OP5	OP6	F1	
SELF REFRESH ENTRY (SRE)	Any	H	L	L	L	H	L	H	H	R1	1, 2, 9
		X	V	V	V	V	V	DSM	PD	F1	
SELF REFRESH EXIT (SRX)	Any	H	L	L	L	H	L	H	L	R1	1, 2
		X	V	V	V	V	V	V	V	F1	
MODE REGISTER WRITE-1 (MRW-1)	Any	H	L	L	L	H	H	L	H	R1	1, 2
		X	MA0	MA1	MA2	MA3	MA4	MA5	MA6	F1	
MODE REGISTER WRITE-2 (MRW-2)	Any	H	L	L	L	H	L	L	OP7	R1	1, 2
		X	OP0	OP1	OP2	OP3	OP4	OP5	OP6	F1	
MODE REGISTER READ (MRR)	Any	H	L	L	L	H	H	L	L	R1	1, 2
		X	MA0	MA1	MA2	MA3	MA4	MA5	MA6	F1	
WRITE FIFO (WFF)	Any	H	L	L	L	L	L	H	H	R1	1, 2
		X	L	L	L	L	L	L	L	F1	
READ FIFO (RFF)	Any	H	L	L	L	L	L	H	L	R1	1, 2
		X	L	L	L	L	L	L	L	F1	
READ DQ CALIBRATION (RDC)	Any	H	L	L	L	L	H	L	H	R1	1, 2
		X	L	L	L	L	L	L	L	F1	
NOTE 1 LPDDR5 commands are one clock cycles long and defined by the states of CS at the rising edge (R1) of clock and CA[6:0] at the rising edge (R1) and the falling edge (F1) of clock. Note that some operations such as ACTIVATE, MODE REGISTER WRITE or WCK2CK SYNC (WS_RD, WS_WR, WS_FS) require two commands to initiate.											
NOTE 2 "V" means "H" or "L" (a defined logic level). "X" means don't care in which case CK_t, CK_c and CA[6:0] can be floated.											
NOTE 3 Bank Group Address BG[1:0] and Bank Address BA[1:0] determine which bank is to be operated upon.											
NOTE 4 An ACTIVATE-1 command must be followed by an ACTIVATE-2 command. Only CAS, WRITE, MASK WRITE, READ, PRECHARGE (to a different bank) commands can be issued between ACTIVATE-1 and ACTIVATE-2 commands. The maximum timing gap between ACTIVATE-1 and ACTIVATE-2 commands are 8 clock cycles. An ACTIVATE-1 command must be issued first before issuing an ACTIVATE-2 command. Once an ACTIVATE-1 command is issued, an ACTIVATE-2 command must be issued before issuing another ACTIVATE-1 command.											
NOTE 5 AB "HIGH" during PRECHARGE or REFRESH command indicates that command must be applied to all banks and bank address is a don't care.											

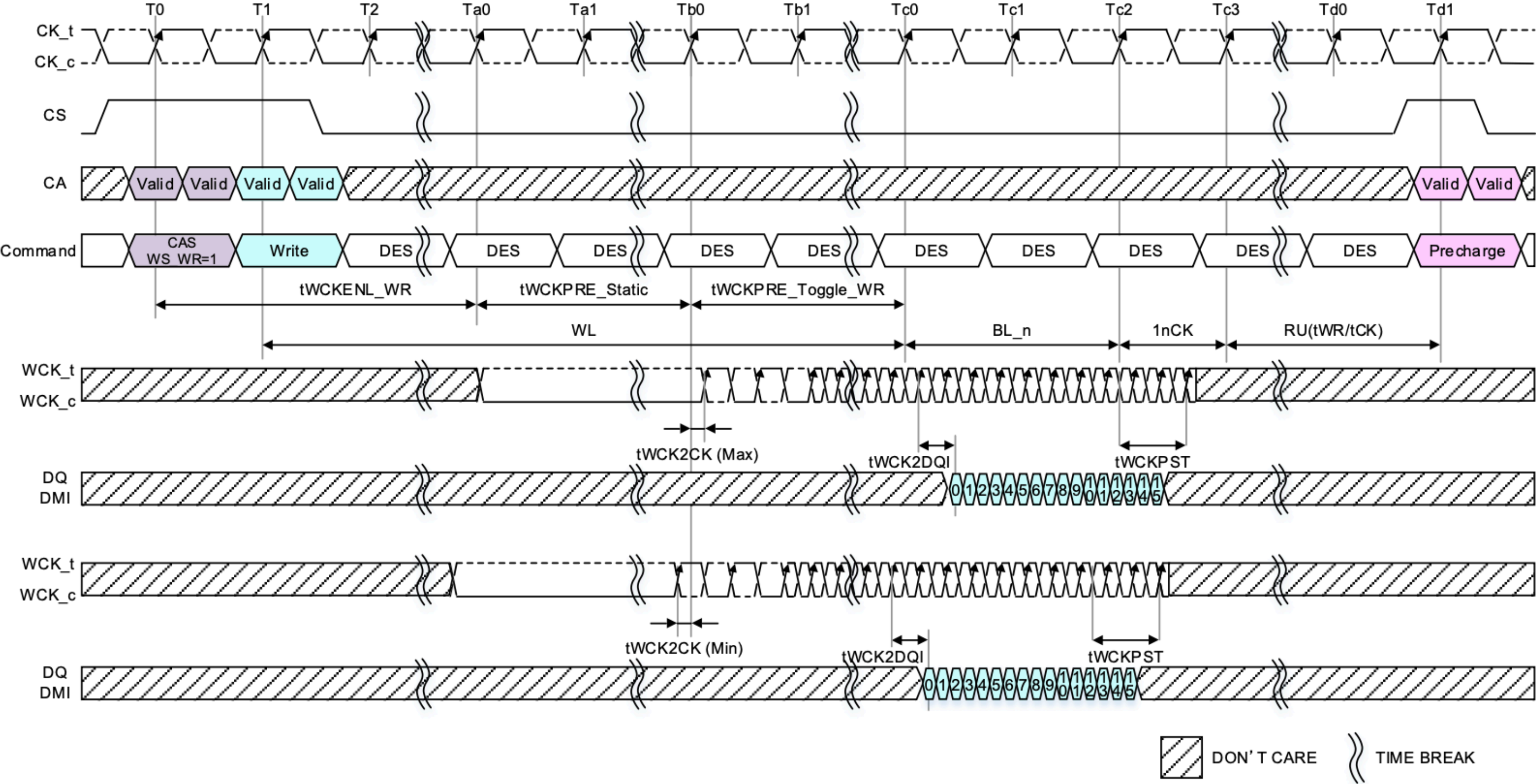
从用户的角度看如何写一个数据进去

Activate



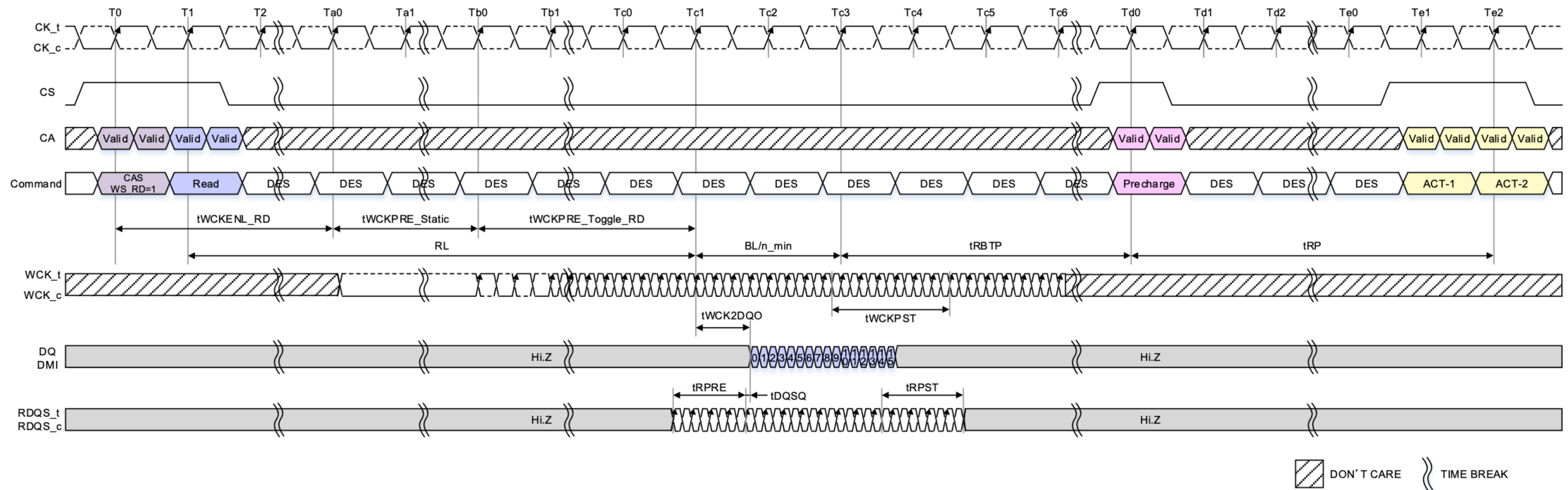
从用户的角度看如何写一个数据进去

Write



从用户的角度看如何读取一个数据回来

Read



上节课后问题解答

- 课后问题1: “DDR指的是数据是双沿的。但是在LPDDR5中的命令也是上下沿采样的”。以上说法是正确的吗? **正确。LPDDR5中的CA也是上下沿采样的。**
- 课后问题2: “Deselect 与 NOP对DRAM产生的作用是相同的, 因此两个命令可以合二为一, 或者说其中一个冗余的”。以上说法是正确的吗? **错误。两个命令对于DRAM产生的作用在某些情况下是不同的。**
- 课后问题3: “Precharge命令与Refresh命令均有all bank和per bank两种模式”。以上说法是正确的吗? **正确。**
- 课后问题5: 在发送Read命令前, 必须保证目标row处于**Active**状态, 且必须保证WCK2CK处于**Sync**状态?
- 课后问题7: 发送Activate命令的前提条件是此时DRAM处于**Idle**状态? 如果不处于此状态, 如何使得DRAM进入此状态? **通过发送Precharge命令**

课后问题

- 课后问题1: WL和RL分别是____和____的缩写?
- 课后问题2: 对于同一个Rank而言, 发送Activate命令到发送Read之间的时间间隔为以下哪一个时序参数? A. tRRD; B. tRCD; C. tRAS; D. tRP; E. tRC
- 课后问题3: 计算Activate的时序参数时, 应当从以下哪个位置作为标记计算? A. ACT1 Rising Edge; B. ACT1 Falling Edge; C. ACT2 Rising Edge; D. ACT2 Falling Edge
- 课后问题4: LPDDR5中CAS命令中与WCK2CK Sync有关的命令有____个? 分别为____? 他们的含义分别是____?
- 课后问题5: “WCK需要与CK完全对齐, 即WCK与CK之间不允许有phase mismatch”。以上说法是正确的吗?
- 课后问题6: 要使得WCK2CK处于Sync状态, 有哪些方法? 要使得WCK2CK处于Unsync状态, 有哪些方法?
- 课后问题7: “与Write操作不同, Read操作时需要的是rdqs, 不需要wck”。以上说法是正确的吗?

下节预告

第六节 - 初步了解DRAM Jedec Spec 4 初始化及Training

- DRAM初始化
- DRAM的基本Training

谢谢